UNIVERSITY OF HYDERABAD

{Establishment by the act of the Parliament in 1974}
Gachibowli, Hyderabad

TENDER SCHEDULE FOR THE WORK OF

MAINTENANCE OF BUILDINGS:-"REPLACEMENT OF 50KW ONLINE SOLAR INVERTERS (02NOS) AND PV MODULES" IN THE UNIVERSITY OF HYDERABAD CAMPUS. NOS)

C O N T R A C T O R

ESTIMATED COST Rs. 7,77,266/- (Inclusive of GST)

EMD Rs. 15,545/-

F & A Receipt No. _____ Book No. _____

Time for the Completion of work: 03 Months

FROM _____TO_____

WORK ORDER REFERANCE:

EXPENDITURE CHARGABLE TO:

Bid Submission start date (Online) : 04.09.2025 up to 3.00 P.M.

Bid Submission close date (Online) : 17.09.2025 up to 3.00 P.M.

Bid opening date (Online) : 18.09.2025 at 3.00 P.M (Technical Bid)

Date of opening of financial bids of technically qualified bidders will be intimated through Central Public Procurement Portal.

Sd/-
Assistant Engineer (E-I)

Sd/-
UNIVERSITY ENGINEER

UNIVERSITY OF HYDERABAD
{Establishment by the act of the Parliament in 1974}
Gachibowli, Hyderabad.
University Works Department,
Ph no.040-23132306/040-23101208
E-mail: aet.uohyd@gmail.com

E-TENDER

FOR

NOTICE INVITING TENDER (NIT)

Notice Inviting e-Tender No.	**UH/ENGG/UWD/2025-26/04**
Name of the work	**Maintenance of Buildings:-**"Replacement of 50KW online solar inverters (02 Nos) and PV Modules" in the University of Hyderabad Campus.
Estimated Cost	**Rs.7,77,266/- (Inclusive of GST)**
Earnest Money Deposit (EMD)	**Rs. 15,545/- (Through DD or Bank Guarantee)**
Performance Guarantee	**5% of the Tendered value i.e 38,863/-**
Time Allowed	**03 Months**
Date of Start of Bid Submission	**04 .09.2025 at 3.00 PM**
Last date & Time for Submission of Bids	**17.09.2025 upto 3.00 PM**
Date & Time of Opening of Technical Bids	**18.09.2025 upto 3.00 PM**
Date & Time of Opening of Finance Bids	**Will be intimated to technically qualified bidders in due course of time through Central Public Procurement portal.**
Submission of hard copies of EMD	The original EMD in favour of Finance Officer, UoH should be submitted in the University Works Department, University of Hyderabad, Gachibowli, P.O. Central University, HYDERABAD – 500046 up to 3.00 PM on 18.09.2025

Sd/-
Assistant Engineer (E-I)

Sd/-
UNIVERSITY ENGINEER

UNIVERSITY OF HYDERABAD

University Works Department

Item Rate Tender Contract for Works

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FORM ‘J’

UNDERTAKING

I/We undertake and confirm that our firm/partnership firm has not been blacklisted by any state/central departments/PSUs/Autonomous bodies and the last 7 years of its operations. Further that, if such information comes to the notice of the department then I/we shall be debarred for bidding in University of Hyderabad in future forever. Also, if such information comes to the notice of University of Hyderabad on any day before date of start of work, the Engineer-in-charge shall be free to cancel the agreement and to forfeit the entire amount of earnest money Deposit/Performance Guarantee.

If further solemnly declare that all the certificates are true to the best of my knowledge.

Signature of Bidder(s) or an authorized Officer of the firm with stamp

Signature of the Contractor

Check list for Technical eligibility of Tender

Sl No	Doc Ref	Description of the Document	Enclosed Yes/No	Remarks
1	NIT Para No 1 Registration of Contractors	CPWD/MES/BSNL/Railway/State PWD (B&R) Autonomous bodies or undertakings.		
2	NIT Para No 1 Eligible Contractors/	Date of validity of enlistment		
3	Details of works completed during last 7 years (Detailed statements to be enclosed)	Not less than 40 % of estimated cost (Three works Electrical installations of Solar related works etc.,)		
		Not less than 50 % of estimated cost (Two works Electrical installations of Solar related works etc.,		
		Not less than 80 % of estimated cost (One work Electrical installations of Solar related works etc.,		
		(Details of works on which contractor meets specified eligibility criteria)		
4	As per the NIT	The tenderer shall have to furnish an undertaking in prescribed format. (form J)		
5	Annual turnover.	30% of the estimated cost.		

Signature of the Contractor

Details of the Contractor

S.No.	Particulars	
1	Name of the Firm	
2	Name of the Contractor	
3	Contractor Details	
(a)	Mobile	
(b)	E-Mail id	
4	Name of the Bank	
5	Name of the Branch	
6	Account No. (prefix with zeros, if any as given on the cheque book/pass book) {please attach cancelled cheque}	
7	IFSC Code of the Branch	
8	MICR Code	
9	PAN No.	
10	GST No.(if applicable)	

Signature of the Contractor

GENERAL GUIDELINES

1. This book of "General Conditions of Contract" is applicable to both types of tenders i.e." Percentage Rate tenders and Item rate tenders". Accordingly, alternative provisions for conditions Nos. 4, 10 & 12 of the General Rules and Directions are given in this book. The appropriate alternatives will be applicable in specific cases depending on whether this is used for percentage rate tender (CPWD-7) or item rate tender (CPWD-8).

2. CPWD-6, Schedules A to F, special conditions/specifications and drawings only will be issued to intending bidders. The HCU-GCC 2016 form will not be issued along with the Tender Documents but the same shall form part of the agreement to be drawn and signed by both parties after acceptance of tender. The HCU-GCC 2016 can be accessed and downloaded from website www.uohyd.ac.in

3. All blanks are confined to Notice Inviting Tender (CPWD-6) and Schedules A to F.

4. Authority approving the Notice Inviting Tenders (NIT) shall fill up all the blanks in CPWD-6 and in Schedules B to F before issue of Tender Papers.

5. The intending bidders will quote their rates in Schedule A (Schedule of quantities in CPWD Form 3)

6. The proforma for registers and Schedules A to F are only for information and guidance. These are not to be filled in the Standard Form. The Schedules with all blanks, duly filled, shall be separately issued to all intending tenderers.

Signature of the Contractor

Method of Tender Application, Tender Submission and Tender Opening

The intending bidder must read the terms and conditions of CPWD-6 carefully. He should only submit his bid if he considers himself eligible and he is in possession of submission of all the required documents as per NIT.

The following Technical and Finance bid documents shall be upload simultaneously with scanned copies in e-procurement web site. The tender documents may also be seen at University of Hyderabad website-www.uohyd.ac.in

(A) <u>Technical bid documents</u>

1. The EMD of 2% of estimated cost should be drawn in favor of "Finance Officer, University of Hyderabad" either in form of fixed deposit receipt of a bank/demand draft/banker check of a scheduled bank. The original EMD shall be placed in Envelope-1 and closed using suitable adhesive and shall be dropped in the Tender box before the bid opening date and time. The scanned copy of EMD shall be uploaded on the web site. Bids submitted along with EMD in the prescribed form shall be considered for further evaluation of eligibility Criteria. Bids submitted without EMD shall be summarily rejected. The Technical Bid of **SSI/MSME/PSU/** (Central & State) **are not exempted** from submission of EMD and transaction fee for tender processing. Purchase preference for **SSI/MSME/PSU** shall not be provided for this tender. The Technical bid shall contain all the documents scanned and uploaded in the concerned wed site, based on which the bidder claim to be eligible as per eligibility criteria prescribed in NIT. Any discloser of finance bid in the technical bid documents will lead to summary rejection of bid.
2. Work experience certificates as per Eligibility criteria.
3. Contract registration certificate registered under State or Central Govt Departments or state/central autonomous bodies undertaking.
4. PAN card & GST Registration Certificate.
5. Latest TDS certificates for consecutive 3 years.
6. Latest 3 years Turn over certificate.
7. Undertaking as per Form "J".
8. Other necessary documents as per Tender documents.

On the due date of opening, the Technical bid of those tender who furnished valid EMD will only be opened. On opening of Technical bid, further detailed scrutiny/evaluation will be carried out. During the evaluation of techno commercial bids, the documents furnished by the tenderers will be scrutinized in detail. Any tender, found as not fulfilling the eligibility criteria will be rejected at this stage and such offers will not be considered for further processing. The price bid of only those tenderers who have been qualified during the scrutiny and technical evaluation will be opened separately on a specified date (with due intimation to the qualified bidders) and further processed, as per tender procedure/stipulations.

Signature of the Contractor

<u>Eligibility Criteria and Notice Inviting tender:</u>

1. Bidder must be a familiar with solar power grid modular integration & Inverters works along with Electrical works for the last 7 years.
2. The bidders must be having similar experience in installation of On grid and OFF grid system.

3. Experience Criteria (Eligibility criteria for participation in tendering).
Estimate Cost for the work is **Rs. 7,77,266/-**

Bidders should have satisfactorily completed Electrical installations of Solar related works for government /semi government / private works during last 7 years ending last day of the month previous to the one in which the tenders are invited.

(i) One Electrical installations of Solar related works etc., completed work not less than **80%** of Estimate cost i.e **Rs. 6,21,813/-**

(or)

(ii) Two Electrical installations of Solar related works etc completed work **each** costing not less **50%** of Estimated cost i.e. **Rs. 3,88,633/-**

(or)

(iii) Three Electrical installations of Solar related works etc completed works each costing not less than **40%** of Estimated cost i.e Rs. **3,10,906/-**

Self-Attested copies of experience certificates are to be submitted.

(b) The value of executed work will be brought to the current costing level by enhancing the actual value of work at simple rate of 7 % per annum to be calculated from the date of completion to the date receipt of tender as mentioned in NIT.

(c) Copy of completion certificates for works issued by private parties shall be supported by TDS (Tax Deducted at Source) certificates.

4. Only work completed and certified by competent authority of the organization will be accepted as experience certificate. Certificate not issued in name of bidder (Contractor) or issued by an authority not competent to issue such certificate will not be accepted.
Mere work order or purchase order etc. issued will not qualify for experience.

5. The copies of relevant certificates of income tax/IT returns, PAN Card, GST register certificated should accompany to the technical bid.

6. To become eligible the tenderer shall have to furnish an undertakings, as per form 'J' of the NIT, Check list and should produce signed pages from page no. 7 to 23 of the Tender Document.

7. The electrical contractor should have 'A' class Electrical license in case it is not having he should produce electrical license holder certificate of supervisor who works on his works.

8. Agreement shall be drawn with the successful tenderer in the prescribed Form No. CPWD 7/8 which is available in the University of Hyderabad website. Tenderer shall quote his rates as per various terms and conditions of the said form which will form part of the agreement.

Signature of the Contractor

9. The time allowed for carrying out the work is **03 months** from the date of start as defined in clause 5 Scheduled "F" or from the first date of handing over of the site, whichever is later, in accordance with phasing if any indicated in the tender document. The site shall be made available as and when required for execution of work.

10. Tenders shall be accompanied with Earnest money of 2% of estimated cost put to tender/fixed deposit receipt of a scheduled bank/demand draft/banker check of a scheduled bank issued in favour of Finance Officer, University of Hyderabad payable at Hyderabad and valid for aa period of 135 days from the date of opening of technical bids.

11. The Bidder/contractor, whose tender is accepted, will be required to furnish performance guarantee of 5% (Five Percent) of the tendered amount (including the EMD amount) within the period specified in Schedule F. This guarantee shall be in the form of cash (in case guarantee amount is less than Rs. 10000/-) or Deposit at Call receipt of any scheduled bank/Banker's cheque of any scheduled bank/Demand Draft of any scheduled bank/Pay order of any scheduled bank (in case guarantee amount is less than Rs. 1,00,000/-) Government Securities or Fixed Deposit Receipts or Guarantee Bonds of any Scheduled Bank in accordance with the prescribed form.
    In case the contractor fails to deposit the said performance guarantee within the period as indicated in Schedule "F" including the extended period if any, the Earnest Money deposited by the contractor shall be forfeited automatically without any notice to the contractor.

12. The description of the work is as follows:
    Copies of other drawing and documents pertaining to the works will be open for inspection by the tenderers at the office of above mentioned officer. Tenderers are advised to inspect and examine the building and its area surroundings and satisfy themselves before submitting their tenders. The tenderer shall be responsible for arranging and maintaining at his own cost all materials, tools & plants, water, required for executing the work unless otherwise specifically electricity access, facilities for workers and all other services provided for in the contract documents. Submission of a tender by a tenderer implies that he has read this notice and all other contract documents and has made himself aware of the scope and specifications of the work to be done.

13. The competent authority on behalf of the University of Hyderabad does not bind itself to accept the lowest or any other tender and reserves to itself the authority to reject any or all the tenders received without the assignment of any reason. All tenders in which any of the prescribed condition is not fulfilled or any condition including that of conditional rebate is put forth by the tenderer shall be summarily rejected.

14. Canvassing whether directly or indirectly, in connection with tenderers is strictly prohibited and the tenders submitted by the contractors who resort to canvassing will be liable to rejection.

15. The University Engineer on behalf of the competent authority of University reserves to himself the right of accepting the whole or any part of the tender and the tenderer shall be bound to perform the same at the rate quoted.

Signature of the Contractor

I

16. No Engineer of Gazette rank or other Gazetted Officer employed in Engineering or Administrative duties in an Engineering Department of the Government of India is allowed to work as a contractor for a period of one year after his retirement from Government service, without the previous permission of the Government of India in writing.

17. This contract is liable to be cancelled if either the contractor or any of his employees is found any time to be such a person who had not obtained the permission of the Government of india as aforesaid before submission of the tender or engagement in the contractor's service.

18. The tender for the works shall remain open for acceptance for a period of ninety (90) days from the date of opening of tenders/Ninety days from the date of opening of financial bid in case tenders are invited on 2 envelop system (strike out as the case may be) if any tenderer withdraws his tender before the said period or issue of letter of acceptance, whichever is earlier, or makes any modifications in the terms and conditions of the tender which are not acceptable to the department, then the Government shall, without prejudice to any other right or remedy, be at liberty to forfeit 50% of the said earnest money as aforesaid. Further the tenderer shall not be allowed to participate in the retendering process of the work.

19. The notice inviting Tender shall form part of the contract document. The successful tenderer/contractor, on acceptance of his tender by the Accepting Authority shall within 15 days from the stipulated date of start of the work, sign the contract consisting of:-

    a. The Notice Inviting Tender, all the documents including additional conditions, specifications and drawings, Clarifications if any, forming the tender as issued at the time of invitation of tender and acceptance thereof together with any correspondence leading thereto.
    b. Standard C.P.W.D. Form 7/8

20. Time and Quality are the essence of this work and to ensure the same each R.A bill shall accompany timely, Quality, Quantity certificate and the same shall be ensured by the University Engineer.

21. GST invoice form should be filled and submitted along with the bill.

22. The labour engaged and working equipment brought by the contractor for the work shall be recorded in the UoH security main gate before entering in to the premises and obtain necessary gate pass before exit.

Signature of the Contractor

Item Rate Tender Contract for Works

A. Tender for the work of: **"Replacement of 50KW online solar inverters (02 Nos) and PV Modules" in the UoH Campus.**

i. To be submitted by _____up to 2.30 P.M. _____.

ii. To be Opened in presence of tenderers who may be present at 3.00 P.M Hyd. On _____in the office of the

Executive Engineer, University Works Department, University of Hyderabad

Issued to ___N/A_____

Signature of officer issuing the tender Documents:

Designation __N/A_____

Date of Issue _____

Signature of the Contractor

TENDER

I/We have read and examined the notice inviting tender, schedule, A, B, C, D, E & F Specifications applicable, Drawings & Designs, General Rules and Directions, Conditions of Contract, clauses of contract, Special conditions, Schedule of Rate & other documents and Rules referred to in the conditions of contract & all other contents in the tender document for the work.

I/We hereby tender for the execution of the work specified for the University of Hyderabad within the time specified in Schedule "F" viz., schedule of quantities and in accordance in all respect with the specifications, designs, drawing and instructions in writing referred to in Rule-1 of General Rules and Directions and in Clause 11 of the Conditions of contract and with such materials as are provided for, by, and in respect of accordance with, such conditions so far as applicable.

We agree to keep the tender open for ninety (90) days from the due date of its opening / ninety days from the date of opening of financial bid in case tenders are invited on 2/3 envelop system (strike out as the case may be) and not to make any modification in its terms and conditions. A sum of Rs._____is hereby forwarded in DD / Bankers Cheque / deposit at call receipt of a scheduled bank/fixed deposit receipt of scheduled bank/demand draft of a scheduled bank issued by a scheduled bank in Favour of 'Finance Officer, UOH' as earnest money. If I/We, fail to furnish the prescribed performance guarantee within prescribed period. I/We agree that the said University of Hyderabad, in office shall without prejudice to any other right or remedy, be at liberty to forfeit the said earnest money absolutely. Further, if I/We fail to commence work as specified, I/We agree that University of Hyderabad in office shall without prejudice to any other right or remedy available in law, be at liberty to forfeit the said earnest money and the performance guarantee absolutely, otherwise the said earnest money shall be retained by him towards security deposit to execute all the works referred to in the tender documents upon the terms & conditions contained or referred to those in excess of that limit at the rates to be determined in accordance with the provision contained in Clause 12.2 and 12.3 of the tender form. Further, I/we agree that in case of forfeiture of earnest money & performance guarantee as aforesaid. I/we shall be debarred for participation in the re-tendering process of the work.

I/We undertake and confirm that eligible similar work(s) has/have not been got executed through another contractor on back to back basis. Further that, if such a violation comes to the notice of Department, then I/we shall be debarred for tendering in University of Hyderabad in future forever. Also, if such a violation comes to the notice of Department before date of start of work, the Project In-Charge shall be free to forfeit the entire amount of Earnest Money Deposit/Performance Guarantee.

Signature of the Contractor

I/We hereby declare that I/We shall treat the tender documents drawings and other records connected with the work as secret/confidential documents & shall not communicate information there from to any person other than a person to whom I/We am/are authorized to communicate the same or use the information in any manner prejudicial to the safety of the State.

Dated:
Signature of Contractor
Witness: Address:
Postal Address:
Occupation:

ACCEPTANCE

The above tender (as modified by you as provided in the letters mentioned hereunder) is accepted by me for an on behalf of the University of Hyderabad for a sum of Rs. _____
(Rupees _____

_____). The letters referred to below shall form part of this contract agreement:-

(a)

(b)

(c)

University Engineer

Signature

……..

Signature of the Contractor

PROFORMA OF SCHEDULES

(Separate Performa for Civil, Elect. & Hort. Works in case of composite Tenders)
(Operative Schedules to be supplied separately to each intending tenderer)

SCHEDULE 'A'

Schedule of quantities (as per CPWD-3)

SCHEDULE 'B'

Schedule of materials to be issued to the contractor.

S.No.	Description of item	Quantity	Rates in figures & words at which the material will be charged to the contractor	Place of issue
1	2	3	4	5
	Nil	Nil	NA	NA

SCHEDULE 'C'

Tools and plants to be hired to the contractor

S.No.	Description	Hire charges per day	Place of Issue
1	2	3	4
	Nil	NA	NA

SCHEDULE 'D'

Extra schedule for specific requirements/document for the work, if Particular Specifications and Tender Conditions attached

Signature of the Contractor

SCHEDULE 'E'

Reference to General Conditions of contract: HCU-GCC-2018

Tender for the work of: "**Replacement of 50KW online solar inverters (02 Nos) and PV Modules" in the UoH Campus.**

Estimated cost of work	:	**Rs.7,77,266/-**
(i) Performance Guarantee	:	5% of tendered value (Including EMD)
(ii) Security Deposit	:	10% of tendered value

SCHEDULE'F'

GENERAL RULES & DIRECTIONS: Officer inviting tender: Engineer-in-Charge/E.E

Maximum percentage for quantity of items of work to be executed beyond which rates are to be determined in accordance with Clauses 12.2 & 12.3:

See below Definitions:

(i) Engineer-in-Charge/E.E is referred to University Engineer or any other Engineer so designated having overall supervision and charge of the execution of the work and who hall sign the contract on behalf of University of Hyderabad.

(ii) Accepting Authority for grant of EOT is the V.C, University of Hyderabad

(iii)	Percentage on cost of materials and Labour to cover all overheads and profits	:	15%
(iv)	Standard Schedule of Rates is DSR	:	2023
(v)	Department	:	U.W.D , UOH, HYDERABAD

(vi) Standard HCU Contract Form GCC 2020, HCU Form 7/8 modified & Corrected up to 2021

(vii) In respect of items of work, for which the contractor has quoted freakishly high rates i.e where the agreement rates are in excess of 50% of the estimated rates/Market rate, agreement rates will be paid up to be agreed quantities only. For the quantities executed in excess of the agreed quantities, payment will be made at the mutually agreed rates as approved by the Competent Authority. In respect of items of work, for which the contractor has quoted rates are below 50% of the estimated rates/Market rate, these rates will be considered as freak low rates. Quantities of these items shall not be decreased under any circumstances.

Signature of the Contractor

(viii) The work shall be carried out according to CPWD Specification, General Condition of contracts 2020 along with special condition/technical specification, drawings applicable. If, however, any particular item/items are issue are not covered by these specifications and special conditions, then the work shall be carried out in accordance to relevant BIS (ISI) codes/code of practice.

(ix) Time is the essence of contract. Accordingly, the contractor shall abide by the time schedule. Failure to complete the work within the stipulated time will entail recovery of compensation from the work bills as determined by the University Engineer, under condition -2 of the General Conditions of contracts.

(x) Warranty of Inverters / PCU is 5 years from the date of completion of the work. 80% of total cost of work against components of Solar power plant against delivery at site with necessary documents will be paid, 15% after Commissioning & Completion will be paid and remaining 5% will be held with UoH as security deposit for 5 years warranty against inverters and can be released only after submission of fixed deposit or Bank Guarantee in favour of Finance Officer University of Hyderabad for a period of 5 years from any scheduled Bank. Security deposit against warranty period will be released only after successful completion of 5 years.

Signature of the Contractor

Clause 1

(i) Time allowed for submission of Performance Guarantee from The date of issue of letter of acceptance . 14.....Days

Clause 2

Authority for fixing compensation under clause 2 is U.E

Clause 2A

Whether Clause 2A shall be applicable No

Clause 5

Number of days from the date of issue of letter of acceptance for reckoning date of start..... 14... Days

Clause 12

12.2. & 12.3Deviation Limit beyond which clauses 12.2 & 12.3 shall apply for building work..... Unlimited

12.5 Deviation Limit beyond which clauses 12.2 & 12.3 shall apply for foundation work ..... Unlimited

Clause 16

Competent Authority for deciding reduced rates. Engineer-in-charge

Clause 18

List of mandatory machinery, tools & plants to be deployed by the contractor at sit

..... 3.
1. .....2.....
5.
4.....6……
.....…….

7. .....8..... 9. .....

Signature of the Contractor

Clause 25

Constitution of Dispute Redressal Committee (DRC)	Competent Authority to appoint DRC
DRC shall constitute of one Chairman and two members	University Engineer

Clause 36 (i)

Requirement of Technical Representative(s) and recovery Rate

Minimum Qualification of technical representative	Discipline	Designation (Principal Technical/ Technical Representative	Minimum Expenditure	Number	Rate at which recovery shall be made from the contractor in the event of not fulfilling provision of clause 36(i)	
					Figures	Words

Assistant Engineers retired from Government services that are holding Diploma will be treated at par with Graduate Engineers.

Clause 42

(i) (a) Schedule/statement for determining theoretical quantity of cement & bitumen on the basis of Delhi Schedule of Rates 2023 printed by C.P.W.D.

(ii) Variations permissible on theoretical quantities:

a. Cement
   For works with estimated cost put in tender not more than Rs. 5 lakh. 3% plus/minus.
   For works with estimated cost put in tender more than Rs.5 lakh. 2% plus/minus.

b. Bitumen All Works 2.5% plus & only & nil on minus side.

c. Steel Reinforcement and structural steel sections for each diameter, section and category 2% plus/minus

d. All other materials. Nil

Signature of the Contractor

Special Conditions

1. Before tendering, the contractor shall inspect the site of work and shall fully acquaint himself about the conditions prevailing at site, availability of materials, availability of land and suitable locations for construction of godowns, stores and camp, transport facilities, the extent of leads and lifts involved in achieving the completion of work.

2. The contractor shall at his own expense and risk arrange land for accommodation of labour, setting up of office, the storage of materials, erection of temporary workshops, and construction of approach roads to the site of the work including land required for carrying out of all jobs connected with the completion of the work. However, the departmental land, to the extent available may be allowed to be used for the purpose free of rent without accepting any responsibility for the delay, if any, on this account. The contractor shall have to abide by the regulations of the authorities concerned and the directions of the Engineer-in-charge for the use of the land available at the site of work. If during construction, it becomes necessary to remove or shift the stored materials shed workshop, access roads, etc, to facilitate execution of any other work by any other agency, the contractor shall carry out the removal of shifting as directed by the Engineer- in – charge and no claim whatsoever, shall be entertained on this account.

3. It shall be deemed that the contractor has satisfied himself as to the nature and location of the work, transport facilities, availability of land for setting up of camp, etc. the department will bear no responsibility for lack of such knowledge & the consequences thereof.

4. The contractor shall have to make approaches to site, if so required and keep them in good condition for transportation of labour and materials as well as inspection of works by the Engineer–in-charge. Nothing extra shall be paid on this account.

5. The Contractor shall at his own cost submit samples of all materials sufficiently in advance and obtain approval of Engineer – in – charge. Subsequently the materials to be used in the actual execution of the work shall strictly conform to the quality of samples approved by the Engineer – in – charge and nothing extra shall be paid on this account. The acceptance of any sample or material on inspection shall not be a bar to its subsequent rejection, if found defective.

6. The contractor shall at his cost, make all arrangements and shall provide necessary facilities as the Engineer – in – charge may require for collecting, preparing, packing, forwarding and transportation of the required number of samples for tests & for analysis at such time and to such places as directed by the Engineer – in - charge. Nothing extra shall be paid for the above operations including the cost of materials required for tests & analysis. Testing charges, if any, will be borne by the department if the test passes. In case it fails, the same shall be borne by the contractor.

7. The necessary tests shall be conducted in the laboratory approval by the Engineer – in-charge. The samples for carrying out all or any of the tests shall be collected by Engineer –in – charge or on his behalf by any other officer of UoH. The contractor or his authorized representative shall associate himself in collection, preparation, packing and forwarding of such samples for the prescribed tests and analysis. In case the contractor or his authorized representative is not present or does not associate himself in the aforesaid operation the results of such tests and consequences thereon shall be binding on the contractor.

Signature of the Contractor

8. Materials used on work without prior approval and testing (where testing is necessary) by the Engineer–in–charge are liable to be considered unauthorized, and not acceptable. The Engineer–in–charge shall have full powers for removal of any or all of the materials brought to site by contractor which are not in accordance with the contract specifications or do not conform, in character or quality to the samples approved by the Engineer–in–charge. In case of default on the part of the contractor in removing rejected materials, the Engineer–in–charge shall be at liberty to have them removed at the risk and cost of the contractor.
9. The contractor shall make his own arrangement of water required for execution of work and get the water tested with regard to its suitability for use in the works and get written approval from the Engineer – in – charge before he proceeds with the use of same for execution of work.
10. The work shall be carried out in such a manner so as not to interfere / or adversely / or disturb other works being executed by other agencies, if any.
11. Any damage done by the contractor to any existing work or work being executed by other agencies shall be made good by him at his own cost.
12. The work shall be carried out in the manner complying in all respect with the requirement of relevant rules and regulations of the local bodies under the jurisdiction of which the work is to be executed and nothing extra shall be paid on this account.
13. The contractor shall maintain in good condition all work executed till the completion of entire work entrusted to the contract under this contract.
14. No payment shall be made to the contractor for damage caused by rain, flood and other natural calamities whatsoever during the execution of works and any damage to the work on this account shall have to be made good by the contractor at his own cost.
15. The rates for all items of work, unless clearly specified otherwise, shall include the cost of all labour materials, de – watering required, if any, and other inputs involved in the execution of the items.
16. Unless otherwise provided in the schedule of quantities, the rates tendered by the contractor shall be all inclusive.
17. No claim whatsoever for idle labour, additional establishments, costs of hire and labour charges for tools and plants etc., would be entertained under any circumstances.
18. For the safety of all labour directly or indirectly employed in the work for the performance of the contractor's part of this agreement, the contractors shall, in addition to the provision of CPWD safety code and directions of the Engineer – in –charge, make all arrangements to provide facility as per the provision of Indian Standard Specifications (Codes) listed below & nothing extra shall be paid on this account.

1.	IS 3696 Part I	Safety Code for scaffolds & ladders
2.	IS 3696 Part II	Safety Code for scaffolds and ladders Part II
3.	IS 764	Safety Code for excavation work.
4.	IS 4138	Safety Code for working in compressed air.
5.	IS7293	Safety Code for working with construction machinery.
6.	IS 7969	Safety Code for storage and handling of building
7.	IS 4130	Safety code for demolition of buildings.

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19. The contractor shall take all precautions to avoid all accidents by exhibiting necessary caution boards and by providing red flags, red lights and barriers. The contractor shall be responsible for any accident at the site of work and consequences thereof.

20. Other agencies working at site will also simultaneously execute the works entrusted to them and to facilitate their working, the contractor shall make necessary provisions

    e.g. holes, openings, etc. for laying / burying pipes, cables, conduits, clamps, hooks, operation for the execution of other agencies smoothly. The final finishing of the work is to be executed in co – ordination with other agencies as directed by the Engineer – in – charge.

21. On account of security considerations, there could be some restrictions on the working hours, movement of vehicles for transportation of materials and location of labour camp. The contractor shall be bound to follow all such restrictions and adjust the programme for execution of work accordingly.

22. In case of any difference in the Hindi version and English version in any of the condition of contract, English version shall prevail.

23. Contractor shall be responsible for the watch and ward of the materials lying at site/building in all respect till the handing over of building.

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GENERAL CONDITIONS

1.1 Scope

The general scope covers the design, details of equipment to be supplied, inspection as may be necessary before dispatch, delivery at site, installation, testing commissioning etc of the Solar PV System.

1.2 Related documents:

These technical specifications shall be read in conjunction with the General conditions of contract with all correction slips, as well as schedules and drawings. In the event of any discrepancy between those specifications and inter-connected contract documents, the technical requirements as per the tender specifications shall be followed and deem to be having over-riding value.

2. Definition of terms:

The definition of terms used in these general specifications are given in the NIT.

3. Conformity with statutory Acts, Rules, Regulations, Standards and Safety Codes:

The installation shall be carried out as per the technical specifications mentioned in the NIT

4. Indian Electricity Act Rules:

All electrical works in connection with installation of electric climatic and comfort chambers shall be carried out in accordance with the provisions of Indian Electricity Act 1910 and the Indian Electricity Rules 1956 amended upto date. The electrical works shall also conform to CPWD General Specifications for Electrical works Part-I(Internal) 2005 and Part-II (External) 1994 amended up to date wherever relevant.

5. Safety Codes and Labour Regulations:

The contractor shall at his own expenses arrange for the safety provisions as per the statutory regulations, IS recommendations, regulations under factory Act etc., where applicable and instructions issued from time to time in respect of all labour employed by him directly or indirectly for the installation of the System.

The contractor shall provide necessary barriers, warning signs and other safety measures etc., wherever necessary so as to avoid accident. In addition all safety procedures as outlined in Appendix 'VI' shall be complied with.

In case of default the department shall be at liberty to make arrangements and provide facilities as aforesaid and recover the cost from the contractor.

He shall also indemnify UoH against claims for compensation arising out of his negligence in this regard.

6. Fire Regulations:

The installation shall be carried out in conformity with the local fire regulations and rules there under wherever they are in force.

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7.0 Works to be done by the Department:

The department shall be responsible for the following works only and these are excluded from the scope of the successful tenderer:

7.1 Provision of 3 phase, 4 wire, 50Hz, 415 volts A.C. Power supply

7.2 Free water and power supply during installation, testing and commissioning periods.

8.0 Works to be done by the contractor:

8.1 In addition to the manufacture, supply, installation, testing and commissioning of the Rooftop solar PV system including all auxiliary equipment, following woks shall be deemed to be included within the scope of the work to be done by the contractor

8.2 All minor building work necessary for installation of equipment such as making of openings in walls/ floors, either of RCC or brick masonry etc. and restoring them to original condition and finish. The scope of minor building work includes all drouting of foundation concrete pad to be formed or made as base for supporting R.S joist etc., grouting and anchoring of all boards clamps, supports foundation bolts, such works shall exclude cutting of partition wall wherever involved.

8.3 All electrical works except bringing in main connection.

8.4 Responsibility to ensure safety of the entire materials against pilferage damage till the installation is handed over to the consignee.

8.5 All scaffolding as may be necessary during erection work and subsequently removed.

8.6 Temporary barricades with caution boards at each landing to prevent accident during execution of work.

8.7 It is the responsibility of the contractor to validate the design parameters specified in actual operating conditions.

9. Coordination with other agencies:

The successful contractor shall coordinate with other contractor/agencies engaged in construction of building if any and exchange freely all technical information so as to make the execution of works contract smooth.

10. Completeness of tender:

All fittings, equipment, unit assemblies and accessories, hardware, foundation bolts, terminal lugs for electrical connections, cable glands, junction box and items which are useful and necessary for efficient assembly in operation and installation shall be deemed to have been included in the scope of work. The installation shall be complete in all details whether such details have been mentioned in the specifications or not.

11. Information to be supplied by contractor after award of work

Within a period of 15 days from the date of receipt of letter of acceptance the contractor shall provide the department his programme bar chart for submission of preliminary drawing, manufacturing of equipment, installation, testing, commissioning and handing over. The contractor shall be required to submit in triplicate the following drawings and information within the above 15 days period for approval of the department before commencing the work.

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11.1 Design of the installation and all general arrangement drawings:

11.2 Details of foundation for equipment, load data location etc., of various assembled equipment as may be needed generally by other agencies for purpose of their work.

11.3 Complete layout dimensions for every unit/group of units with dimensions required for erection purposes.

11.4 Any other drawing/information not specifically mentioned above but deemed to be necessary for the job by the contractor.

11.5 List of items to be carried out by the department in accordance with the tender accepted.

12. Commencement of work

As soon as the preliminary drawings are approved, the contractor should commence work. The contractor shall also send three sets of final drawings to the department who shall return one copy.

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TECHNICAL SPECIFICATIONS

1. Contractors are advised to visit the site before quoting the rates. Otherwise it will be assumed that the party has already visited the site before quoting the tender.
2. Power Conditioning Unit (PCU): PCU of minimum 50 KVA (+/- 5%) Central and modular type, should convert DC power produced by SPV modules, in to AC power and adjust the voltage & frequency levels to suit the local grid conditions. Inverter shall interconnect and feed power to the grid at 415V. Mandatory Technical Specification is as below:

2.1. **The technical specification of PCU is listed below:**

String inverter - PVS-50/60-TL		
Technical data and types		
Type code	**PVS-50-TL**	**PVS-60-TL**
Absolute maximum DC input voltage (Vmax,abs)		1000 V
Start-up DC input voltage (Vstart)	420…700 V (Default 420 V)	420…700 V (Default 500 V)
Operating DC input voltage range (Vdcmin...Vdcmax)	0,7 x Vstart ..950 V (min 300 V)	0,7xVstart ...950 V (min 360 V)
Rated DC input voltage (Vdcr)	610 Vdc	720 Vdc
Rated DC input power (Pdcr)	52000 W	61800 W
Number of independent MPPT		3 (version SX and SX2) / 1 (versione standard e S)
Maximum DC input power for each MPPT (PMPPT, max)	19300W@30°C / 17500W@45°C	23100W@30°C / 21000W@45°C
MPPT input DC voltage range (VMPPTmin ... VMPPTmax) at Pacr	480-800 Vdc	570-800 Vdc
Maximum DC input current (Idcmax) for each MPPT		36 A
Maximum input short circuit current for each MPPT		55 A (165 A in case of parallel MPPT)
Number of DC input pairs for each MPPT		5 (SX and SX2 versions), 1 (standard and S version)
DC connection type		Screw terminal block (Standard and -S version) or PV quick fit connector 1) (-SX and SX2 version)
Input protection		
Reverse polarity protection Yes, from limited current source	Yes, from limited current source	
Input over voltage protection for each MPPT Type 2 / Type 1 + 2 (option)	Type 2 / Type 1 + 2 (option)	
Photovoltaic array isolation control According to local standard	According to local standard	

DC switch rating for each MPPT (version with DC switch) 75 A / 1000 V for each MPPT	75 A / 1000 V for each MPPT	
Fuse rating (version with fuses) / max fuse rating 15A (1000V) / 20A (1000V)	15A (1000V) / 20A (1000V)	
Output side		
AC grid connection type	Three-phase (3Ph/N/PE or 3Ph/PE), grounded WYE system only	
Rated AC power (Pacr @cosf=1)	50000 W	60000 W
Maximum AC output power (Pacmax @cosf=1)	55000 W up to 30°C 2)	66000 W up to 30°C 2)
Maximum apparent power (Smax)	55000 VA up to 30°C 2)	66000 VA up to 30°C 2)
Rated AC grid voltage (Vac,r)	400 V	480 V
AC voltage range	320...480 V 3)	384...571 V 3)
Maximum AC output current (Iac,max)		80A
Contributory fault current		92A
Rated output frequency (fr)		50Hz/60Hz
Output frequency range (fmin...fmax)		47..53Hz/57..63Hz
Nominal power factor and adjustable range		> 0.995; 0...1 inductive/capacitive with maximum Sn
Total current harmonic distortion		<3%
Maximum AC cable		95mm2 copper or stranded aluminum
AC connection type		Screw terminal block, cable gland (admitted cable diameter 25..44mm)
Output Protection		
Anti-islanding protection		According to local standard
Maximum external AC overcurrent protection		100A
Output overvoltage protection		Type 2
Operating performance		
Maximum efficiency (hmax)	98.3%	98.50%
Euro efficiency	98%	98%
Communication		
Embedded communication interfaces	3x RS-485, 2X Ethernet (RJ45), WLAN (IEEE802.11 b/g/n @ 2,4 GHz)	
Communication protocol	Modbus RTU / TCP (Sunspec compliant); Aurora Protocol	
Remote monitoring services	Standard level access to Aurora Vision monitoring portal	
Advanced features	Integrated Web User Interface; Embedded logging and direct transferring of data to Cloud	
Technical data and types		
Type code	PVS-50-TL	PVS-60-TL
Environmental		
Ambient temperature range	-25...+60°C (-13...140 °F) with derating above 45 °C (113 °F) with derating above 45 °C (113 °F)	
Relative humidity	4%... 100% condensing	

Sound Pressure level, typical	75 dB(A) @1 m	
Maximum operating attitude	4000 m (13123 ft) with derating above 2000 m / 6561 ft	
Physical		
Environmental protection rating	IP65	
Cooling	Forced air	
Dimension (HXWXD)	68 Kg / 150 ibs (S X version)	
Mounting System	Single mounting bracket	
Safety		
Marking	CE	
Safety and EMC standard	IEC/EN 62109-1, IEC/EN 62109-2, EN 61000-6-2, EN 61000-6-3, EN 61000-3-11, EN 61000-3-12 EN 62311, EN 301 489-1, EN 301 489-17, EN 300 328	
Grid standard (check your sales channel for availability)	CEI 0-21, CEI 0-16, DIN V VDE V 0126-1-1, VDE-AR-N 4105, G59/3, DRRG/DEWA, Chile LV/MV EN 50438, RD 1565, RD 413, UTE C15-7-712-1 P.O. 12.3, AS/NZS 4777.3, BDEW, NRS-097-2-1, MEA, PEA, IEC 61727, ISO/IEC Guide 67(System 5) IEC 61683, VFR-2014, IEC 62116, Synergrid C10/11, IRR-DCC-MV, CLC-TS-50549-1/-2, G99, EN 50549-1/-2	
Available Product variants		
Input connections with terminal blocks + surge arrester Type 2 in both DC and AC sides	PVS-50-TL	PVS-60-TL
Input connections with terminal blocks + DC switch + surge arrester Type 2 in both DC and AC sides	PVS-50-TL-S	PVS-60-TL-S
15 quick Input connections + fuses (single pole) + DC switch + surge arresters Type 2 in both DC and AC sides	PVS-50-TL-SX	PVS-60-TL-SX
15 quick Input connections + fuses (both poles) + DC switch + surge arresters Type 2 in both DC and AC sides	PVS-50-TL-SX2	PVS-60-TL-SX2
Optional available		
SPD Type 1+2 on the DC side	Available	Available
Display	Available	Available
Negative Grounding kit	PVS-50/60-Grounding Kit	PVC-50/60 GROUNDING KIT

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3. Other important Features/Protections required in the INVERTER

3.1 Automatic morning wake-up and night shutdown

3.2 PCU must have the feature to work in tandem with other similar PCUs and be able to be successively switched ON and OFF automatically based on solar radiation variations during the day.

3.3 Mains (Grid) over-under voltage and frequency protection Fool proof protection against ISLANDING. Included authentic tracking of the solar array‘s maximum power operation voltage (MPPT).

3.4 Array ground fault detection.

3.5 LCD and piezoelectric keypad operator interface Menu driven Automatic fault conditions reset for all parameters like voltage, frequency and/or black out.

3.6 MOV type surge arresters on AC and DC terminals for over voltage protection from lightning-induced surges.

3.7 INVERTER should be rated to operate at 0–55 deg. Centigrade unless provision for air conditioning is included in INVERTER

3.8 All parameters should be accessible through an industry standard communication link.

3.9 The INVERTER shall be self commuted and shall utilize a circuit topology and Components suitable for meeting the specifications listed above at high conversion efficiency and with high reliability.

3.10 Since the INVERTER is to be used in solar photo voltaic energy system, it should have high operational efficiency. The DC to AC conversion efficiency shall at least be 98 percent for output ranging from 20 percent of full load to full load. The idling current at no load must not exceed 2 percent of the full-load current.

3.11 In INVERTER there shall be a direct current isolation provided at the output by means of a suitable isolating transformer.

3.12 The INVERTER output shall be 415 V AC, 50 Hz 3 phase supply.

3.13 The INVERTER shall be capable of operating in parallel with the grid utility service and shall be capable of interrupting line-to-line fault currents and line-to-ground fault currents.

3.14 The INVERTER shall include appropriate self-protective and self-diagnostic features to protect itself and the PV array from damage in the event of INVERTER component failure or from parameters beyond the INVERTER‘s safe operating range due to internal or external causes. The self-protective features shall not allow signals from the INVERTER front panel to cause the INVERTER to be operated in a manner which may be unsafe or damaging. Faults due to malfunctioning within the INVERTER, including commutation failure, shall be cleared by the INVERTER protective devices and not by the existing site utility grid service circuit breaker.

3.15 The INVERTER shall go to shut down/standby mode, with its contacts open, under the following conditions before attempting an automatic restart after an appropriate time delay; in sufficient solar power output.

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3.16 **Insufficient Solar Power Input.**

3.16.1 When the power available from the PV array is insufficient to supply the losses of The INVERTER, the INVERTER shall go to a standby/shutdown mode. The INVERTER control shall prevent excessive cycling during rightly shut down or extended periods of insufficient solar radiation.

3.17 **Utility-Grid Over or Under Voltage**

3.17.1 The INVERTER shall restart after an over or under voltage shutdown when the utility grid voltage has returned to within limits for a minimum of two minutes.

3.18 **Utility-Grid Over or Under Frequency**

3.18.1 The INVERTER shall restart after an over or under frequency shutdown when the utility grid voltage has returned to the within limits for minimum of two minutes.

3.19 The INVERTER generated harmonics measures at the point of connection to the utility services when operating at the rated power shall not exceed a total harmonic current distortion of 3 percent, a single frequency current distortion of 4 percent and single frequency voltage distortion of 1 percent, when the first through the fiftieth integer harmonics of 50 Hz are considered.

3.20 The INVERTER Power factor at the point of utility service connection shall be >0.99 lagging or leading when operating at above 25 percent of the rated output, but may be less than 0.99 lagging below 25 percent of the rated output.

3.21 The high voltage and power circuits of the INVERTER shall be separated from the low-voltage and control circuits. The internal copper wiring of the INVERTER shall have flame resistant insulation. Use of PVC is not acceptable. All conductors shall be made of standard copper.

3.22 The INVERTER shall withstand a high voltage test of 2000 V rms, between either the input or the output terminals and the cabinet (chassis).

3.23 Full protection against accidental open circuit and reverse polarity at the input shall be provided.

3.24 The INVERTER shall not produce Electromagnetic Interference (EMI) which may Cause malfunctioning of electronic and electrical instruments including Communication equipment, which are located within the facility in which the INVERTER is housed.

3.25 The INVERTER shall have an appropriate display on the front panel to display the instantaneous AC power output and the DC voltage, current and power input. Each of these measurement displays shall have an accuracy of 1 percent of full scale or better. The display shall be visible from outside the INVERTER enclosure. Operational status of the INVERTER, alarms, trouble indicators and ac and the dc disconnect switch positions shall also be communicated by appropriate messages or indicator lights on the front cover of the INVERTER enclosure.

3.26 Communication Modbus protocol with LAN/WAN options along with remote access facility and SCADA package with latest monitoring systems including string monitoring with Web/IP data monitoring shall be provided.

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3.27 **Electrical safety, earthing and protection**

3.27.1 Internal Faults: In built protection for internal faults including excess temperature, commutation failure, overload and cooling fan failure (if fitted) is obligatory.

3.27.2 Galvanic Isolation: Galvanic Isolation is required to avoid any DC component being injected into the grid and the potential for AC components appearing at the array.

3.27.3 Over Voltage Protection: Over Voltage Protection against atmospheric lightning discharge to the PV array is required. Protection is to be provided against voltage fluctuations in the grid itself and internal faults in the power conditioner, operational errors and switching transients.

3.27.4 Earth fault supervision: An integrated earth fault device shall have to be provided to detect eventual earth fault on DC side and shall send message to the supervisory system.

3.27.5 Cabling practice: Cable connections must be made using PVC Cu cables, as per BIS standards. All cable connections must be made using suitable terminations for effective contact. The XLPO DC Grade Cu cables must be run in GI trays with covers for protection.

3.27.6 Fast acting semiconductor type current limiting fuses at the main bus-bar to protect from the grid short circuit contribution**.**

3.28 The INVERTER shall include an easily accessible emergency OFF button located at an appropriate position on the unit.

3.29 The INVERTER shall include ground lugs for equipment and PV array grouting. The DC circuit ground shall be a solid single point ground connection in accordance with WEC 69042.

3.30 All exposed surfaces of ferrous parts shall be thoroughly cleaned, primed, and painted or otherwise suitably protected to survive a nominal 30 years design life of the unit.

3.31 The INVERTER enclosure shall be weatherproof and capable of surviving **climatic changes and should keep the INVERTER** intact under all conditions in the room where it will be housed. **The INVERTER shall be located indoor and should be either wall/ pad mounted**. Moisture condensation and entry of rodents and insects shall be prevented in the INVERTER enclosure. The enclosure for housing the inverter shall be minimum IP54 protection level. The inverter itself shall be minimum IP20 protection level.

3.32 Components and circuit boards mounted inside the enclosures shall be clearly identified with appropriate permanent designations, which shall also serve to identify the items on the supplied drawings.

3.33 All doors, covers, panels and cable exists shall be gasketed or otherwise designed to limit the entry of dust and moisture. All doors shall be equipped with locks. All openings shall be provided with grills or screens with openings no larger than 0.95 cm. (about 3x8 inch).

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3.34 In the design and fabrication of the INVERTER the site temperature (5 to 55 C), incident sunlight and the effect of ambient temperature on component life shall be considered carefully and derating shall be applied. Similar consideration shall be given to the heat sinking and thermal for blocking diodes and similar components.

3.35 **Factory Testing**: The INVERTER shall be tested to demonstrate operation of its Control system and the ability to be automatically synchronized and connected in parallel with a utility service, prior to its shipment.

Operation of all controls, protective and instrumentation circuits shall be demonstrated by direct test if feasible or by simulation operation conditions for all parameters that cannot be directly tested.

Special attention shall be given to demonstration of utility service interface protection circuits and functions, including calibration and functional trip tests of faults and isolation protection equipment.

Operation of startup, disconnect and shutdown controls shall also be tested and demonstrated. Stable operation of the INVERTER and response to control signals shall also be tested and demonstrated.

Factory testing shall not only be limited to measurement of Phase currents, efficiencies, harmonic content and power factor, but shall also include all other necessary tests/simulation required and requested by the Departments Engineers. Tests may be performed at 25, 50, 75 and 100 percent of the rated nominal power. A Factory Test Report (FTR) shall be supplied with the unit after all tests. The FTR shall include detailed description of all parameters tested qualified and warranted. Factory testing of the INVERTER should be carried out and witnessed by the Department‘s Engineers at the manufacturers premises.

3.36 **Operating Modes:**

3.36.1 The following operating modes are to be made available: Night or Sleep mode: Where the inverter is almost completely turned off, with just the timer and Control system still in operation, losses should not exceed 1 watt per 5kilowatt.

3.36.2 **Standby mode** : Where the control system continuously monitors the output of the solar generator until pre-set value is exceeded (typically 10 watts) Operational or MPPT tracking mode: The control system continuously adjust the voltage of the generator to optimize the power available. The power conditioner must automatically re-enter stand- by mode when input power reduces below the standby mode threshold. Front Panel display should prove the status of the INVERTER, including AC Voltage, CurrentPower output & DC Current, Voltage and Power input, pf and fault Indication (if any).

3.37 **Codes and Standards**

3.37.1 The quality of equipment supplied shall be controlled to meet the guidelines for engineering design included in the standards and codes listed in the relevant ISI and other standards, such as:

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3.37.2 IEEE 928 Recommended Criteria for Terrestrial PV Power Systems. IEEE 929 Recommended Practice for Utility Interface of Residential and Intermediate PV Systems.

3.37.3 IEEE 519 Guide for Harmonic Control and Reactive Compensation of Static Power Controllers.

3.37.4 National Electrical NEPA 70-(USA) or equivalent national standard. National Electrical Safety Code ANSI C2- (USA) or equivalent national standard.

3.37.5 JRC Specification 503 (Version 2.2 March 1991) or JPL Block V standard for PV modules.

3.37.6 The inverter manufacturer should attach efficiency certificate from Independent Third party Testing laboratory i.e. IEC,TUV,SNL/ERTL & STQC. PCU should confirm to IEC 61683 IS 61683 for efficiency measurements and IEC 60068 (1,2,14,30)/Eqvt BIS Std. for environmental testing. MPPT unit should confirm to design qualification IEC 62093.

3.38 PLANT METERING / DATA LOGGING

3.38.1 PV array energy production: Digital Meters to log the actual value of AC/ DC Voltage, Current One way LT energy meter (Export metering) Class 0.2S ABT

3.38.2 Compliant shall be incorporated in the system on the main AC Grid supply complete with electrical switchgear in the main plant.

3.38.3 Solar Irradiance: An integrating Pyranometer (Class I) should be provided, with the sensor mounted in the plane of the array. Readout should be integrated with data logging system.

3.38.4 Wind Speed: An integrated wind speed measurement unit to be provided.

3.38.5 Temperature Sensor: Integrated temperature sensors for measuring the module surface temperature, inverter inside enclosure temperature and ambient temperature to be provided complete with readouts integrated with the data logging system.

3.38.6 A data logging system (Hardware and software) for plant control and monitoring shall be provided with the following features :

3.38.7 Remote Supervisory Control and data acquisition through SCADA software at the Departments location through Handheld device / GSM cellular device with latest software/ hardware configuration and service connectivity for online / real time data monitoring/ control complete to be supplied and operation and maintenance/control to be ensured by the Contractor.

3.38.8 All major parameters should be available on the digital bus and logging facility for energy auditing through the internal microprocessor and can be read on the digital front panel at any time the current values, previous values for up to a month and the average values. The following parameters should be accessible via the operating interface display. AC Voltage, AC Output current, Output Power DC Input, Voltage DC Input, Current Time Active, Time Disabled Time Idle, Temperatures Inverter and Status

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3.38.9 Protective function limits (Viz-AC Over voltage, AC Under voltage, Over frequency, Under frequency ground fault, PV starting voltage, PV stopping voltage, Over voltage relay, Under voltage relay over frequency, ground fault relay, PV starting relay, stopping relay).

3.39 **INVERTER/ARRAY SIZE RATIO**

The Inverter continuous power rating shall be above 98% at full load.

3.40 **Inverter**

3.40.1 Technical data sheet:- The inverter shall work continuously and control the utility interface within the stipulated range.

On three Phase side:-

Output Voltage	415 (+12.5-20%) VAC
Frequency	50H(+3HZ,3HZ)
Maximum current ripple	3% PP
Reactive Power	0.99 inductive to 0.99 capacitive

3.41 **Maximum Power Point Tracker (MPPT)**

Maximum power point tracker shall be integrated in the Inverter to maximize energy drawn from the array. The MPPT should be micro processor based to minimize power losses. The details of working mechanism of MPPT shall be mentioned. The MPPT must have provision (manual setting) for constant voltage operation.

3.42 **Disconnection and Islanding**

3.42.1 Disconnection of the PV generator in the event of loss of the main grid supply is to be achieved by in built protection within the power conditioner. This may be achieved through rate of change of current, Phase angle, unbalanced voltage or reactive load variants.

3.42.2 Operation outside the limits of power quality as described in the technical data sheet should cause the power conditioner to disconnect the grid. Additional parameters requiring automatic disconnection are:

Neutral voltage displacement over current, Earth fault and reverse power

3.42.3 In case of the above, cases, tripping time should be less than 0.5 seconds. Response time in case of grid failure due to switch off or failure based shut down should be well within 5 seconds. In case of use of two or more than two Inverters capacity, suitable equipment for synchronizing the AC output of both the Inverters to the ACDB / Grid should be provided.

3.43 Automatic reconnection after the grid failure is restored

3.42.4 INVERTER shall have facility to reconnect the inverter automatically to the grid following restoration of grid, subsequent to grid failure condition.

The system should have integrated SCADA and software for plant control and remote communication with web monitoring to monitor individual strings and complete power plant.

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4. ARRAY JUNCTION BOX WITH STRING MONITORING FEATURE TO THE INVERTER

4.1 The junction boxes are to be provided in the PV yard for termination of connecting cables. The Junction Boxes (JB) shall be made of FRP/Powder Coated Aluminum with full dust, water & vermin proof arrangement. All wires/cables must be through cable lugs. The JBs shall be such that input & output termination can be made through suitable cable glands.

4.2 Made of FRP or cast aluminium Copper bus bars/terminal blocks housed in the junction box with suitable termination threads Conforming to IP65 standards and IEC 62208Hinged door with EPDM rubber gasket to prevent water entry. Single compression cable glands. Provision of earthing Suitable capacity MOVs provided within the box to protect against lightning

4.3 Plant Control, data logger & plant monitoring unit Basically, this unit should perform the following.

4.4 Individual Array monitoring via string monitoring system Measurement and/or recording of energy parameters.

4.5 Simple data logger or energy meter to record the energy data on a pre-determined interval basis.

4.6 Measurement & continuous acquisition of ambient air temperature, wind speed, solar radiation, PV module temperature, individual string current, inverter output voltage and current, output frequency

4.7 Operating state monitoring and failure indication.

4.8 Representation of monitored data's in graphics mode or in tabulation mode. Controlling & monitoring the entire power system through remote terminal as well as from a local terminal.

4.9 Necessary hardware's & software's shall have to be supplied by the contractor. Both the software's and hardware's required for interfacing the plant with SPS office including CPUs, modems, Printers, UPS, Cellular device are to be supplied and installed by the contractor as per Clause 3.27.

4.10 Remote control/ Instrumentation: The microprocessor control unit should have the provision for installation of RS – 232/485 communication link, should remote control and monitoring capability (by personal computer) be desired. All parameters, status and indicators and targets accessible through the local operator interface may be accessed remotely through these ports. Optional analog outputs (0-5VDC) for AC powers, DC current, DC Voltage can be supplied to interface with external data acquisition systems. Optional contacts inputs from an external SCAD/RTU or other remote control device can be provided within the inverter enclosure for remotely disabling or resetting the unit.

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5. AC DISTRIBUTION PANEL BOARD

5.1 AC Distribution Panel Board (DPB) shall control the AC power from inverter, and should have necessary surge arrestors. Interconnection from ACDB to LT bus be carried out and complete equipment along with metering to be installed in the ACDB. Requirement/specifications of DCDB and ACDB may be changed as per site conditions. The AC panel shall be provided with adequate safety features to prevent transmission of fault.

5.2 All switches at the circuit breakers, connectors should conform to IEC 60947 Part I, II and III.

6. CABLES & WIRES

6.1 Cabling in the yard and control room: Cabling in the yard shall be carried out as per IE Rules. All other cabling above ground should be suitably mounted on cable trays with proper covers. Only LSHZ XLPO cables must be used for DC side, DC grade cables shall be used.

6.2 Wires: Only FRLS copper wires of appropriate size and of reputed make shall be used.

6.3 Cable Ends: All connections are to be made through suitable cable / lug / terminals, crimped properly & with use of Cable Glands.

6.4 Cable Marking: All cable/wires are to be marked in proper manner by good quality ferule or by other means so that the cable can be easily identified.

6.5 Any change in cabling schedule/sizes if desired by the Contractor/Contractor be got approved after citing appropriate reasons. All cable schedules/layout drawings have to be got approved from the Department prior to installation. All cable tests and measurement methods should confirm to IEC 60189.

7. LIGHTNING AND OVER VOLTAGE PROTECTION

There shall be the required number of suitable lightning arrestors installed in the array field. Lightning protection shall be provided by the use of early seamer (Piezo electric) and suitable earthing such that induced transients find an alternate route to earth. Protection shall meet the safety rules as per Indian Electricity Act. Lighting protection for array yard and control room building has to be provided. All building earth conductors shall be interconnected through the concept of earth mats for interconnection with separate earth pits. For each earth pits necessary test points shall have to be provided.

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8. EARTHING PROTECTION

Each array structure of the PV yard should be grounded properly. In addition the lightning arrester / masts should also be provided inside the array field. Provision should be kept be provided inside the array field. Provision should be kept for shorting and grounding of the PV array at the time of maintenance work. All metal casing/shielding of the plant should be thoroughly grounded in accordance with Indian Electricity Act 2003 / Indian Electricity Rules 1956. Earth Resistance should be tested in presence of the representative of UoH after earthing by calibrated earth tester. INVERTERS, ACDB and DCDB should also be earthed properly. The MV equipment's and parts shall be earthed in compliance to Indian Electricity Rules 1956.

Rules 1956, all noncurrent carrying metal parts shall be earthed with two separate and distinct earth continuity conductors to an efficient earth electrode.

9. TOOLS & TACKLES AND SPARES:

9.1 After completion of installation & commissioning of the power plant, necessary tools & tackles are to be provided free of cost by the contractor for maintenance purpose. List of tools and tackles to be supplied by the contractor for approval of specifications and make from UoH.

9.2 A list of requisite spares in case of INVERTER comprising of a set of control logic cards, IGBT driver cards etc, Junction Boxes, Fuses, MCCBs etc. along with spare set of PV modules and batteries be indicated, which shall be supplied along with the equipment. A minimum set of spares shall be maintained in the plant itself for the entire period of warranty and Operation & Maintenance which upon its use shall be replenished. A minimum spare parts list is enclosed at Annexure 1 in Section IX.

10. Plant Electricals

10.1 Solar Photovoltaic Power Plant Electrical System

The technical requirements of design, manufacture, testing at works supply installation testing & commissioning of all electrical equipment's required for the Solar Photovoltaic Power Plant starting from the local control panel of Plant and up to the Grid tie up with the State grid including all control protection, metering equipment, and balance of equipments complete in all respect shall be of high standard and quality meeting the requirement of respective Indian Standards. The brief particulars and requirement of equipment is as under-

10.2 Electricity Rules Act & Rules:

All work shall be carried out in accordance with the latest edition of the Indian Electricity Act and rules formed there under and as amended from time to time

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10.3 Indicating and Integrating Meters / Instruments:

All indicating instruments shall be of switchboard type, back connected, suitable for flush mounting and provided with dust and vermin proof cases for tropical use and finished in suitable colour. All instruments shall have practical laboratory means for adjustment of accuracy. The limits of errors for ammeters/voltmeters shall be those permissible for Class 1.5 instruments as per IS:1248. The ammeters and voltmeters shall be suitably scaled to indicate the current / voltage levels in the system. A phase selector switch with four/six position shall be used to measure the current / voltage of each phase / line. The meters shall be located at eye level to facilitate observation of readings correctly.

10.4 TEMPERATURE RISE:

The maximum temperature rise of various parts of the system when tested under rated conditions shall not exceed the specified values at a peak ambient temperature of 40 deg. C.

10.5 INSPECTION AND TESTING:

Each equipment shall comply with and shall be subjected to all routine and acceptance tests prescribed in the relevant Indian Standard Specification/IEC. The type test reports of switchgear shall be complete in all respect along with oscillographic records, photographs etc. in respect of all type tests as per relevant ISS / IEC. The type test certificates should be in respect of specific make and type / rating of the instruments etc. intended to be supplied and not in respect of the breakers etc. manufactured by their foreign collaborators if any. Complete literature of switchgear proposed to be provided must be sent with the tender.

Routine & acceptance test as per relevant standard shall be carried out on each equipment covered by this specification in the presence of Department's representative. If so desired by the Department all test reports shall be submitted and got approved from the Department before dispatch of the equipment.

11. TECHNICAL PARTICULARS :

The Contractors shall furnish the guaranteed technical & other particulars of the equipments offered. Make of various equipments should be clearly stated. Words like reputed, equivalent etc. shall not be accepted. Alternative makes of equipments should not be more than two in the order of preference.

12. INTERCHANGEABILITY:

All similar materials and removable parts of similar equipments shall be interchangeable with each other.

13. FAULT / TROUBLE ALARM SCHEME:

All non-trip alarms shall be indicated by an alarm bell.

Make / type of each relay, indicating instruments, integrating instruments, control switches, selector switches, indicating lamps, semaphore indicators, enunciator scheme, bell, hooter etc. shall be clearly and invariably indicated in the GTP (Guaranteed Technical Particulars), bill of material and unit price list. Only specific make accessories shall be indicated. The word "EQUIVALENT/REPUTED MAKE" will not be given for consideration.

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14. Guaranteed Technical Particulars, Drawings And Quality Assurance Plan:

14.1 Guaranteed Technical Particulars in the Technical Particulars Data Sheet- with this specification shall be furnished along with the detailed general arrangement dimensional drawings mounting arrangements, connection diagrams and quality assurance plan etc.

14.2 Particulars which are subject to guarantee shall be clearly marked in GTP.

Name Plate and Marking:

The equipment shall have a non-detachable type name plate which should be clearly visible and effectively secured against removal having markings as per requirement of IS. A sticker reading as under must be provided in the MV neutral compartment.

CAUTION: — DO NOT REMOVE EARTH LINK WHEN MV TERMINAL IS LIVE.

14.3 Earthing system installation shall be in strict accordance with the latest editions of Indian Electricity Rules, relevant Indian Standards and code of practices and the local statutory authority regulations.

Neutral points of system metallic enclosures and frame works, not forming part of electric supply shall be connected to main earthing system.

14.4 **Earthing Layout:**

The contractor shall submit to UoH earthing drawings showing the location of earthing conductors, for their approval. Earthing conductors in outdoor areas shall be buried 600 mm below finished graded level and these buried conductors shall be brought 500 mm above ground level for making tap connections to the equipments.

15. Specification For Testing & Commissioning

The testing and commissioning for all electrical equipment at site shall be according to the procedures listed below:

15.1 All electrical equipment shall be tested, installed and commissioned in accordance with the latest relevant standards and code of practices published by Indian Standards Institution wherever applicable and stipulations made in relevant general specifications**.**

15.2 The testing of all electrical equipment as well as the system as a whole shall be carried out to ensure that the equipment and its components are in satisfactory condition and will successfully perform its functional operations. The inspection of the equipment shall be carried out to ensure that all materials, workmanship and installations conform to the accepted design, engineering and construction standards, as well as accepted code of practices and stipulations made in the relevant general specifications.

15.3 The contractor in the presence of UoH shall carry out all tests/ Consultant using his own calibrated instruments, testing equipment as well as qualified testing personnel.

15.4 The results of all tests shall conform to the specification requirements as well as any specific performance data, guaranteed during finalization of the contract.

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16. Preparation of the Equipment for Commissioning;

16.1 After completion of the installation at site and for the preparation of plant commissioning, the contractor shall check all the equipment and installation in accordance with the agreed standards, latest relevant code of practices of Indian Standards and specific instructions furnished by the particular equipment manufacturer as well as the department.

16.2 Checking required to be made on all equipment and installations at site shall comprise, but not limited to, the following:

16.3 Physical inspection for removal of any foreign bodies, external defects, such as damaged insulators, loose connecting bolts, loose foundation bolts etc.
Check for grease, insulating / lubricating oil leakage and its proper quantity. Check for the free movement of rotating parts of the rotating machines and devices.

16.4 Check for tightness of all cable joints and bus bar termination ends as well as earth connections in the main earthing network.

16.5 Check for clearance of live bus bars and connectors from the metal enclosure. Check for proper alignment of all draw out devices, if any.

16.6 Continuity checks in case of power and control cables.
Check for proper connection to earth network of all non-current carrying parts of the equipment and installation.

16.7 The relevant tests shall be carried out in accordance with relevant IS of latest issue. The tests which are to be carried out on the equipment shall include, but not be limited to, the following:

16.8 Check for completeness of installation. each pole to earth insulation resistance test.

16.9 Cables Insulation resistance test with 1,000 V megger for cables rated up to 1.1kV grade.

16.10 All 1.1 kV cables shall be subjected to high voltage test after joining and terminating but before commissioning as per relevant standards.
In each test, the metallic sheath / screen / armour should be connected to earth.

16.11 Continuity of all the cores, correctness of all connections as per wiring diagram, correctness of polarity and phasing of power cables and proper earth connection of cable glands, cable boxes, armour and metallic sheath, shall be checked.

16.12 **Earthing System**

16.12.1 Tests to ensure continuity of all earth connections.

16.12.2 Tests to obtain earth resistance of the complete network by using earth tester. The test values obtained shall be within the limits**.**

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17. Lightning & Over Voltage Protection

17.1 The SPV Power Plant should be provided with Lightning and over voltage protection connected to proper earth mats. The main aim of over voltage protection is to reduce the over voltage to a tolerable level before it reaches the PV or other sub-system components. The source of over voltage can be lightning or other atmospheric disturbance.

17.2 The lightning Masts /Conductors shall be made as per applicable Indian standard / International Standard in order to protect the entire Array Yard/Shed from Lightning stroke. Necessary concrete foundation for holding the lightning conductor, in position, to be made after giving due consideration to maximum wind speed and maintenance requirement at site in future.

17.3 The lightning masts / conductor shall be earthed through flats and connected to the Earth mats as per applicable Indian Standards with earth pits. Each Lightning Conductor shall be fitted with individual earth pit as per required Standards including accessories, and providing masonry enclosure with cast iron cover plate having locking arrangement, watering pipe using charcoal or coke and salt as per required provisions of IS.

17.4 The Contractor shall ensure adequate lightning protection to provide and acceptable degree of protection as per IS for the array yard. If necessary more numbers of Lightning conductors may be provided. Design calculations and detailed explanations (in 4 sets) shall be provided for approval of UoH within 15 days of order.

17.5 Lightning Conductor shall be as per requirements of IS Standards.

17.6 All building earth conductor shall be interconnected through the concept of Earth Mats for interconnection with separate earth pits.

17.7 For each earth pit, necessary Test Point shall have to be provided.

18. Earthing System

18.1 The earthing for LT side (Solar Power Plant side) array and LT power system shall be as required as per provisions of Indian Standard/International Standard. Necessary provision shall be made for bolted isolating joints of each earthing pit for periodic checking of earth resistance.

18.2 Each array structure of the SPV Yard/shed shall be grounded properly. The array structure are to be connected to earth pits as per IS standards.

18.3 The earthing for the power plant equipment shall be made as per provisions of IS. Necessary provisions shall be made for bolted isolating joints of each earthing pit for periodic checking of earth resistance**.**

18.4 The complete earthing system shall be mechanically & electrically connected to provide independent return to earth. All three phase equipment shall have two distinct earth connections.

18.5 An Earth Bus shall be provided inside the control room.

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18.6 For each earth pit, necessary Test Point shall have to be provided.

18.7 In compliance to Rule 33 and 61 of Indian Electricity Rules, 1956 (as amended up to date) all noncurrent carrying metal parts shall be earthed with two separate and distinct earth continuity conductors to an efficient earth electrode.

18.8 Earth resistance of the earth pits shall be tested in presence of the representative of UoH

19. Cables & Installation accessories LT cables:

1.1kV Grade, Al. Conductor XLPE insulated Armoured Cables shall be used for all LT Power Cables between Power & Motor Control Cubicles, MCC, Respective feeders, etc. These cables shall be laid in trents or and using Galvanized Cable trays of adequate strength. The cable shall be terminated using Al. Lugs of adequate cross section area. The PMCC‘s and the MCC‘s shall be located in the Basement with cable entry on top cables on DC side shall be XLPO and DC Grade.

19.1 Control Cables:

19.1.1 1.1kV Grade, Cu. Conductor, XLPE insulated Armoured Cables shall be used for all control cables required for the Solar Power Plant. These cables shall be laid on structural supports and using Galvanized Cable trays of adequate strength. The cable shall be terminated using Cu. Lugs of adequate cross section area.

19.1.2 Cu. Conductor, XLPE insulated Armoured with miller insulation between each pair and tinned copper screening. All cables shall be PVC insulated with appropriate grade conforming to IS.

19.1.3 The wiring for module inters connection shall be with hard PVC conduit of approved make. All Tees, Bends etc., shall be approved make. Before procurement, approval for materials should be obtained from RSMML

19.1.4 Only terminal cable joints shall be accepted. No cable joints to join two cable ends shall be accepted.

19.1.5 Cables inside the control room shall be laid in suitable Cable Trays of approved type.

19.1.6 All wires used on the LT side shall conform to IS and should be of appropriate voltage grade. Only copper conductor wires of reputed make shall be used.

19.1.7 Cable terminations shall be made with suitable cable lugs & sockets etc., crimped properly and passed through brass compression type cable glands at the entry and exit point of the cubicles. The panels‘ bottoms should be properly sealed to prevent entry of snakes/lizard etc. inside the panel.

19.1.8 All cable/wires shall be marked with good quality letter and number ferrules of proper sizes so that the cables can be identified easily. The terminal end of cables and number ferrules of proper sizes so that the cables can be identified easily.

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20. Painting & Finish:

20.1 All metal surfaces shall be thoroughly cleaned of rust, scale, oil, grease, dirt etc. Fabricated structures shall be pickled and then rinsed to remove any trace of acid. The under surface shall be made free from all imperfections before undertaking the finishing coat.

20.2 After Phosphate treatment, two (2) coats of yellow zinc chromate primer shall be applied followed by two (2) coats of epoxy based synthetic enameled paint. Shade shall be Siemens Grey Ral- 7032. Thickness of paint shall be not less than 75 micron.

20.3 All unpainted steel parts shall be cadmium plated or suitably treated to prevent rust formation. If these parts are moving elements then they shall be greased.

21. Inspection & Testing

21.1 UoH shall have free access to Contractor's/ manufacturer's works to inspect, expedite and witness shop tests. Any materials or work found to be defective or which does not meet the requirements of the specification will be rejected and shall be replaced at Contractor's cost. Department reserves the right to carry out stage wise inspection of fabrication and components. The Contractor shall furnish a detailed quality assurance plan for UoH review.

21.2 The test & inspection shall be carried out at manufacturer's work and at Plant site with the Contractors obligation. The test and Inspection shall be done in accordance with the relevant standards and the Manufacturer's standard before the delivery to site as well as after the erection and commission at site. The Contractors shall give the list of tests that they will carry out at site to show the performance of plant.

21.3 The data of each test and inspection shall be recorded and submitted as soon as the test/ trials are conducted and will also be a part of final documentation. The shop test shall be carried out to prove the performance parameters of the offered model. The testing shall be done in the presence of the representatives of the department.

21.4 The department shall be sending its 2 representatives for inspection of stage manufacturing and testing at works & 7 days training at manufacturer's works of SPV module and PCU manufacturer. Expenses on account of to &fro journey during stay shall be borne by department. The notice of such inspection shall be given 15 days in advance.

21.5 Manufacturer has to submit procedure for tests carried out at their factory premises.

a. Start Up Trials Load Test
b. Records & Measurements Safety Device List Setting values for all sensors for Pressure and Temperature.
c. Dimensional Check-up, Overall Inspection, Completeness of Scope of Supply Shop Test/Load Test for Solar Power Plant
d. Load Trials & Reliability test at site for Grid Connect Solar Power Plant.

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e. Performance Guarantee Test at Site for Grid Connect Solar Power Plant etc. These tests will be conducted at site as per site conditions at available load and after performing all pre-commissioning check and trials and after readiness of the entire Solar Power Plant system which are required to carry out the load trials.

f. All the tests which are mentioned in the load test of Solar Power Plant will be carried out in presence of UoH Representative at Plant site at site conditions and the parameters checked in accordance with the data sheet and guaranteed parameters given by the Contractor.

g. All the equipment supplied by the vendor will be tested as per relevant standard/ Quality assurance plan at site conditions and the performance monitored.

22. Quality Issues

22.1 Contractor will submit and get finalized detailed comprehensive Standard Field Quality Plan (SFQP) within 30 days from date of issue of the order for bought out items and items manufactured by them. The Standard Field Quality Plan shall relate to the specific and objective erection practices right from storage of equipment till final inspection and testing to be followed for bought out items and items manufactured by Contractor. Accordingly, the Manufacturing Quality Plan shall be submitted broadly under following sub-heads:-

22.2 Raw material/Bought Out items and Components.

22.3 In process inspection and test/checks to establish successful completion/accomplishment of the process.

22.4 Final tests/checks in accordance with relevant national / international standards / specifications.

22.5 The quantum of check for each and every inspection/test items shall be based on an established sampling method and the quantum of check indicated in the SFQP should be designed adequate quality protection.

22.6 In case reference documents/acceptance norms are indicated as per plant standards then the same shall be duly substantiated/properly explained by well-established and proven engineering practices. All submissions will be in English language only.

22.7 Contractor will allow UoH to carry out Quality/Audit/Quality surveillance on Contractors and our sub vendor‘s work with reference to contractual obligations to ensure that the quality management practices/norms as detailed out in the Quality Manual are adhered to. To facilitate this activity, bidder shall keep UoH informed all progress of work in this contract on monthly basis.

22.8 Contractor will associate/fully witness in each inspection being carried out at their/their sub-vendor‘s works by our authorized inspection engineer(s).

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22.9 UoH shall also carry out quality audit and quality surveillance of your systems, procedures and quality control activities. However, this shall not relive you of any of your contractual responsibilities under the contract.

23. OTHER CIVIL STRUCTURES

23.1 All Civil shall be executed as per General Specification of Civil Works. Scope: The scope of work includes:

23.2 Array structures Embedment of structures suitable for mounting of PV modules on terrace/ on elevated array mounting structure/ on the roof of semi-permanent shed and laying of earthing equipment/structures and connecting to the main ground mat as per statutory requirements.

23.3 Providing foundation for lightning arrestors etc. co-ordinating with BESCOM for getting approval for Solar Power Plant and net metering. Providing of pedestals required for mounting of the inverters and control panels.

23.4 All structural drawings to be got approved from UoH.

Tests :All material to be used for construction and fabrication should be tested before use. All laboratory/Field tests should be conducted as per relevant Indian standards and CPWD specifications by taking minimum required samples.

23.5 **DANGER NOTICE BOARDS**

Danger notice boards should be provided as and where necessary as per IE Act / IE rules as amended up to date.

24. DRAWINGS & MANUALS

24.1 5 copies of Engineering, electrical drawings and installation and O&M manuals are to be supplied to the department.

24.2 Contractors shall provide complete technical data sheets for each equipment giving details of the specifications along with make/makes in their bid along with basic design of the power plant and power evacuation, synchronization and distribution along with protection equipment. Approved ISI and reputed makes for equipment be used.

24.3 For complete electro-mechanical works, contractors shall supply complete design, details and drawings for approval to UoH before progressing with the installation work.

25. QUALITY AND ADAPTABILITY OF THE EQUIPMENT

25.1 Contractors must verify the grid behavior, solar insulation levels and general site conditions on their own before bidding. The Contractor shall accordingly ensure that the equipment and the design submitted shall be able to perform as per guaranteed performance levels in the available site conditions. The design of the plant and the equipment offered by the Contractors shall be evaluated for its quality and adaptability to the site conditions based on the Departments past experience, projects earlier executed by the Contractors and from other sources. Contractors must submit detailed technical operational parameters and latest plant performance indicators and status complete with the clients address and contact nos. with regard to projects of similar nature already executed.

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26. SURGE PROTECTOR AT STAGE II / CLASS C

26.1 The surge Protection manufacturer shall offer a complete line of surge Protection product to support the requirements for the Distribution. The surge protector at this stage shall be provided to protect the downstream electrical and electronics against any induced switching surges that may be passed on to the downstream electrical & electronic system.

26.2 The Protection unit shall be based on Single High Capacity Metal Oxide Varistors (MOV), capable of handling 8/20 µs surges and shall be able to give an indication in the event module failure and be pluggable to facilitate the in-service replacement without distributing the lines. One extra set of replacement module shall be furnished to the job site.

26.3 Protection Network Configuration. The work required under this section consists of furnishing, installing and connecting SPD device as specified and as shown in the drawings. The SPD device shall be installed in a NETWORK configuration, consist of one set of SPD panel device at the service entrance of switchboard. All SPD devices in this network configuration shall be of same manufacturer. All SPD devices shall be modular, mountable on 35 mm DIN rail and be field replaceable without interruption of electrical distribution circuit.

26.4 Unit status indicator shall be provided to indicate the status of complete Protection unit on the product as well as provision for remote indication must be provided.

26.5 Protection shall be manufactured for the specific type and voltage of the electrical Service and shall provide clamping for both normal (L-N) and common (N-G) mode operation.

26.6 Protection shall be manufactured to withstand a maximum continuous operating voltage of not less than 115% of normal RMS Line voltage of 240 VAC.

26.7 The Protection shall be provided with internal safety fusing if required, to be connected in parallel between Line/s to neutral & neutral to earth as per the TNS configuration of wiring. It shall be testable on line for routine maintenance, module failure and in order to prevent catastrophic failure modes.

26.8 Protection shall be a fail-safe type device, shall have no follow through current shall have repeated surge capability, shall be solid state, shall be self-restoring and be fully automatic in all mode of operation. It shall have thermal disconnection and indication against overloading of the device.

26.9 Protection shall comply with IEC 61643 standards.
Protection shall have an operating temperature ranges from -20oC to + 60oC.

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26.10 Protection Criteria

The maximum continuous operating voltage (Rated voltage) for SPD devices connected to phase-neutral shall not be less than the values as shown in table below:

Nominal Voltage Rating per phase	**Maximum Continuous Operating Voltage**
(V rms)	(V rms)
120	150
240	320
350	440
480	600

27. Listing

27.1 The surge protective device and associated hardware must comply with IEC 61643-11.

27.2 The Protection voltage of the complete rail mount surge protective device shall be type test to the figures as indicated in table below, which must not exceed the values shown.

Service Voltage / per phase	**Protection Voltage @ In (Nominal discharge current) / Protection Level**
240 V	1500 V

28. Nominal Withstand Surge Current.

28.1 Surge Protective device (including all fusing and over current protection) for application at sub-Distribution Panels shall have a Nominal surge current withstand capacity as shown in table below. The failure or operation of any fuse / over – current device during the test is not permissible.

Application Panel Location	**Max. Single Withstand Surge Current Of 8/20 µs Impulse)**
Sub-Distribution Panel Final Distribution Board	10KA for 8 / 20 µs (between Line to Neutral) 25 KA for 10/350 µs (between Neutral to earth

28.2 Compliance to this specification must be provided in the form of a certificate from an independent testing laboratory. Response time of Class C arrestor should not be <25 ns.

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LIST OF PREFERRED MAKES OF MATERIALS

Solar PV Plant

S. No.	Details of Materials / Equipment	Manufacturer's Name
	A: SOLAR EQUIPMENT	
1.	SPV Modules (subject to compliance with tender conditions)	HHV/ REC/ Canadian Solar/ TATA / BHEL /Panasonic/ Emvee/Shan
2.	SOLAR CELLS	Solar Tech/ Motec/ Q Cell/ Eton
3.	Power Conditioning Unit (PCU)	Preferred **ABB** /SMA / Schneider/ Refusol/ Kaco/ Siemens/Delta
4.	Data logger/System performance	As per OEM/ Engineer-in-charge
5.	Junction Box	Hensel/ Spelsberg/ ABB/ Cooper busman/ Equipment
6.	ACDB panel	Local fabrication approved Engineer- in-charge/OEM
7.	Energy meter	L&T/ Siemens/ Schneider/ Secure/ HPL
8	Surge protection	OBO Betterman/ ABB/ Legrand/ Equ
9.	PV module interconnecting Connector	Tyco/ Cooper busman/ MC
10.	AC cables and wires	RR Kabel/ Havells / Finolex/ Asmon/ Anchor/ KEI/polycab/ Siechem/ LAPP/ TopSolar/ Nexans/ Equ
11.	DC cables and wires	
12.	PVC Conduits and accessories (ISI marked)	Avon Plast/ Precision/ AKG/ Bajaj plast/ GM
13.	GI Pipes	TATA/ JINDAL (HISSAR)/ ZENITH
14.	Ligtning arrestor	INdelec/ Star/ Erico
15	Indicating Lamps LED type , Push Button	Schneider Electric (MG)/ Larsen & Toubro (ESBEE)/ Siemens/ Vaishno Conzerv/ Larsen & Toubro/ Secure
16.	Electronic Digital Meters Networking type)	

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S. No.	Details of Materials / Equipment	Manufacturer's Name
17.	PVC insulated XLPE aluminium/ copper conductor armoured MV Cables upto 3300V grade	CCI/ Havells/ Gloster/ Universal
18.	Selector Switch, Toggle switch	Salzer (Larsen & Toubro)/ Siemens/ Kaycee
19.	Timer	Siemens/ L&T/ Schneider Electric – TE/ GE/ Siemens
20.	Welding Rod	Advani / Esab
21.	Fire Sealant	Promat/ Hilti/ Birla – 3 M
22.	Cat-6 LAN cable	D-link/Finolex/ Systimax/ D-G Link
23.	1.1KV grade aluminium & copper conductor, XLPE insulated armoured UG Cables	FINOLEX/ GLOSTER / HAVELLS / RR CABLE/Polycab
24.	MCCB	Schneider Electric/L & T/Legrand/ABB

Note:-The Engineer-in-Charge, UoH reserves the right to add or delete any materials and makes/brands in the list of preferred makes of materials/brands.

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NAME OF WORK:-"Maintenance of Buildings: "Replacement of 50KW online solar inverters (02 Nos) and PV Modules" in the University of Hyderabad Campus.

1) The estimates amount is Inclusive of GST.
2) The bidder should quote basic rates exclusive of GST and quote GST percentage separately; GST should be revised from time to time.

S.No	Description of items	Qty	Unit	Basic Rate in Figures to be entered by the Bidder without GST	Add: GST (%)	Total Amount without GST	Total Amount with GST
1	Supply, Installation, testing and commissioning of 50kW Solar system a) on grid Power Conditioning Unit (PCU) / Solar inverter of 420-700 V DC Input Voltage range and 400 V AC, Three phase, 4 wire, 50 Hz+/- 2.5 Hz, output voltage suitable to generate AC power with efficiency not less than 97%, total harmonic distortion less than 3%, with IP65 protection rating, having 15 quick Input connections + fuses (both poles) + DC switch + surge arresters Type 2 in both DC and AC sides and suitable for ambient temperature form 0 to 50 degree C. b) Data Monitoring system complete with accessories. c) The PCU shall adjust the Voltage and frequency level to suit the Grid Voltage Frequency, and termination arrangement for incoming and outgoing cable along with glands, lugs and other accessories etc as required.	2.00	Nos				
2	Supply ,Transportation and installation of 320Wp PV Modules	10.00	Nos				

3	Dismantling of Inverter	2.00	Nos				
4	Dismantling of existing Solar PV Modules and removing from UoH Campus	10.00	Nos				
	Total Amount						

Signature of the Contractor	**Sd/-** **University Engineer**	**Sd/-** **Assistant Engineer (E-I)**	**Sd/-** **D'M Gr-I**